VLSM IP Address Design and Dynamic Network Implementation Using DHCP

This project focuses on designing an efficient IP addressing scheme using Variable Length Subnet Masking (VLSM) and implementing dynamic IP address assignment using DHCP for Bahir Dar Institute of Technology (BIT).

1. Number of Networks Required

The institute requires separate networks for each department and one WAN network. Therefore, a total of five (5) networks are required.

2. VLSM Design and Feasibility

The given network is 10.11.48.0/26 which provides 62 usable host addresses. Since the Computing Department alone requires 60 hosts and other departments also require IP addresses, the /26 network is insufficient. Hence, the network is extended to 10.11.48.0/25 to properly apply VLSM and meet all requirements.

3. VLSM Subnet Calculation

Department	Network Address	CIDR	Subnet Mask	Usable Range	Broadcast	Usable Hosts
Computing	10.11.48.0	/26	255.255.255.192	10.11.48.1 – 10.11.48.62	10.11.48.63	62
Electrical	10.11.48.64	/27	255.255.255.224	10.11.48.65 – 10.11.48.94	10.11.48.95	30
Mechanical	10.11.48.96	/28	255.255.255.240	10.11.48.97 – 10.11.48.110	10.11.48.111	14
Material Science	10.11.48.112	/29	255.255.255.248	10.11.48.113 – 10.11.48.118	10.11.48.119	6
WAN	10.11.48.120	/30	255.255.255.252	10.11.48.121 – 10.11.48.122	10.11.48.123	2

Question 4: Subnet Allocation Order Using VLSM

Question 4: Subnet allocation must start from the largest host requirement to the smallest.

According to Variable Length Subnet Masking (VLSM) principles, subnets are allocated starting from the department with the largest number of required hosts to the smallest. This approach minimizes IP address wastage and ensures efficient utilization of the network.

Allocation Order	Department	Hosts Required	Subnet Size
1st (Largest)	Computing Department	60	/26
2nd	Electrical Department	30	/27
3rd	Mechanical Department	14	/28
4th	Material Science Department	6	/29
5th (Smallest)	WAN (Point-to-Point)	2	/30

This allocation order strictly follows the requirement to assign subnets from the largest to the smallest host demand, ensuring proper VLSM implementation.

5. Network Implementation Using Cisco Packet Tracer

The complete network topology is implemented using Cisco Packet Tracer. Router interfaces and WAN links are configured with static IP addresses, while end devices obtain IP addresses dynamically using DHCP pools configured on the router.

6. Documentation and Verification

The documentation includes the VLSM calculation table, DHCP pool configurations, network topology diagram, and screenshots showing successful DHCP assignment and connectivity tests such as ping between devices.